

Quantium Retail Analytics – Task 2 Report

Experimentation and Uplift Testing Analysis

Prepared for: Quantum Virtual Internship

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Executive Summary

This report evaluates the effectiveness of the new store layout trial implemented across three trial stores (77, 86, and 88). The objective of the analysis was to determine whether the new layout resulted in a statistically significant increase in sales performance compared with carefully selected control stores.

Control stores were selected based on historical similarity in sales and customer purchasing behaviour during the pre-trial period using correlation and magnitude distance metrics. Statistical uplift testing was then performed to compare trial and control stores throughout the trial period.

The results indicate that Stores **77** and **88** experienced significant improvements in sales performance, whereas Store **86** did not demonstrate a statistically significant uplift despite an increase in customer traffic.

Overall, the trial can be considered successful, with recommendations to roll out the new layout across additional stores while conducting further investigation into Store 86.

1. Business Objective

The Category Manager requested an evaluation of the new chip category layout trial to determine whether the redesigned store layout positively influenced customer purchasing behaviour and increased total sales.

The specific objectives were:

- Identify appropriate control stores for each trial store.
 - Compare trial store performance against matched control stores.
 - Measure the statistical significance of any observed sales uplift.
 - Identify the key drivers behind sales changes.
 - Provide business recommendations regarding future rollout.
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2. Dataset Overview

The analysis was conducted using the Quantum retail transaction dataset containing historical customer purchases across multiple stores.

The dataset includes:

- Store Number
- Transaction Date
- Loyalty Card Number
- Transaction ID
- Product Quantity
- Total Sales
- Product Information

Monthly performance metrics were generated for each store to facilitate comparison during the pre-trial and trial periods.

3. Key Performance Metrics

The following monthly KPIs were calculated for every store:

Total Sales Revenue

Total dollar sales generated by each store every month.

Number of Customers

Unique loyalty card holders making purchases.

Total Transactions

Number of completed transactions.

Average Transactions per Customer

Measures shopping frequency.

$$\text{Average Transactions} = \frac{\text{Transactions}}{\text{Customers}}$$

Average Chips Purchased per Customer

Measures purchasing quantity.

$$\text{Average Chips} = \frac{\text{Total Quantity}}{\text{Customers}}$$

Average Price per Unit

Measures average selling price.

$$\text{Average Price} = \frac{\text{Total Sales}}{\text{Total Quantity}}$$

4. Methodology

The analysis consisted of four major phases.

Phase 1 – Data Preparation

The transaction dataset was cleaned and transformed into monthly store-level summaries.

This included:

- Date conversion
 - Monthly aggregation
 - KPI calculation
 - Removal of incomplete stores
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Phase 2 – Control Store Selection

Selecting an appropriate control store is essential for measuring the true effect of the trial.

Each trial store was compared with every other store using:

Pearson Correlation

Measures similarity in monthly sales trends.

Higher correlation indicates similar performance patterns.

Magnitude Distance

Measures how close the actual sales values are.

Similarity Score:

$$Similarity = 1 - \frac{Observed - Min}{Max - Min}$$

The final similarity score combined:

- Sales correlation
- Customer correlation
- Sales magnitude
- Customer magnitude

The highest overall score was selected as the control store.

5. Selected Control Stores

Trial Store Control Store

77	233
86	155
88	237

These stores demonstrated the highest similarity during the pre-trial period.

6. Trial Assessment

Store 77

Sales Performance

The trial store consistently outperformed its control store during the trial period.

Statistical testing indicated significant positive sales uplift in multiple trial months.

Customer Performance

Customer numbers also increased significantly compared with the control store.

Business Interpretation

The increase in sales was primarily driven by attracting more purchasing customers rather than changes in transaction frequency.

Store 86

Sales Performance

Sales during the trial period were not significantly different from the matched control store.

Customer Performance

Customer numbers increased slightly but failed to generate meaningful revenue growth.

Business Interpretation

The store experienced higher customer traffic; however, purchasing behaviour did not translate into increased sales.

Additional investigation is recommended.

Store 88

Sales Performance

Store 88 exhibited strong sales growth throughout the trial period.

The observed uplift exceeded expected variation based on historical performance.

Customer Performance

Customer count also increased significantly.

Business Interpretation

The trial layout successfully attracted additional customers and increased total sales.

7. Statistical Testing

A statistical uplift analysis was performed by:

- Scaling control store performance
- Calculating percentage differences
- Estimating pre-trial standard deviation
- Computing t-statistics
- Comparing against the 95% confidence interval

Decision Rule:

If

$$t > t_{critical}$$

then the uplift is considered statistically significant.

8. Key Findings

Store 77

- ✓ Significant increase in sales
 - ✓ Significant increase in customers
 - ✓ Trial successful
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Store 86

- ✓ Customer growth observed
 - ✗ Sales uplift not statistically significant
 - ✓ Further investigation recommended
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Store 88

- ✓ Significant increase in sales

✓ Significant increase in customers

✓ Trial successful

9. Business Insights

The analysis indicates that customer acquisition was the primary driver of increased sales in successful trial stores.

The redesigned layout appears to improve product visibility and customer engagement, resulting in higher purchasing activity.

Store 86 suggests that layout changes alone may not guarantee improved sales, highlighting the importance of considering additional factors such as promotional activity, inventory availability, and local customer demographics.

10. Recommendations

Based on the statistical analysis and business findings, the following recommendations are proposed:

- Roll out the new store layout to additional stores based on the positive performance observed in Stores 77 and 88.
 - Conduct a detailed investigation into Store 86 to identify operational or market-specific factors limiting sales performance.
 - Continue monitoring customer purchasing behaviour following implementation to validate long-term effectiveness.
 - Extend future trials across a larger sample of stores to improve confidence in rollout decisions.
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11. Conclusion

This analysis successfully evaluated the effectiveness of the store layout trial using control store matching and statistical uplift testing.

The trial demonstrated measurable improvements in sales and customer acquisition for two of the three trial stores, providing strong evidence that the new layout positively influences customer purchasing behaviour.

The findings support a broader rollout of the new layout while recommending additional analysis for Store 86 before full implementation.

Overall, the experiment achieved its objective and delivered actionable, data-driven insights that can support strategic retail decision-making.

Appendix

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- SciPy

Statistical Techniques

- Pearson Correlation
- Magnitude Distance Similarity
- Control Store Matching
- Uplift Testing
- Confidence Interval Analysis
- Hypothesis Testing

Project Deliverables

- Cleaned Dataset
- Monthly KPI Tables
- Trial Assessment Visualizations
- Statistical Analysis
- Business Report
- GitHub Documentation